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Circles That Mutate

How to detect genetic change beyond chromosomes

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Abstract

Extrachromosomal circular DNA (eccDNA) and its large oncogenic counterpart, ecDNA, are pervasive yet systematically overlooked components of the somatic genome. Unlike linear chromosomes, circular DNA lacks telomeres, replicates via error-prone rolling-circle mechanisms, and segregates randomly during mitosis. These properties confer an accelerated mutational dynamics upon the circular compartment, yet standard genomic workflows—from DNA extraction and exome capture to short-read alignment and variant calling—are optimized for linear, diploid reference architectures. This review synthesizes current knowledge of circular DNA mutagenesis across five domains. First, the mutational spectrum of circular DNA is distinct from that of chromosomal DNA, enriched for microhomology-mediated indels, complex structural rearrangements, and interchromosomal chimeric fusions. Second, experimental isolation methods, including exonuclease digestion and rolling circle amplification, introduce specific biases that distort mutational discovery. Third, computational detection requires graph-based assembly, junction-aware realignment, and copy-number-normalized burden estimation to distinguish true somatic variants from amplification artifacts. Fourth, validation strategies—inverse PCR, mutation-specific FISH, and single-cell sequencing—confirm that ecDNA mutations exhibit transient inheritance yet drive sustained phenotypic effects through temporary transcriptional dominance. Fifth, the absence of a universal circular reference genome, the difficulty of discriminating replication errors from authentic mutations, and the lack of clinical standardization remain critical barriers to translation. Despite these challenges, the integration of ecDNA analysis into oncogenomic panels is urgently warranted: ecDNA amplifications are present in approximately 14% of all cancers, confer poor prognosis independent of chromosomal copy number, and serve as privileged sites for the acquisition of therapy resistance mutations. Furthermore, eccDNA mutation burden correlates with cellular age and genotoxic stress, suggesting utility as a quantitative biomarker. Emerging therapeutic strategies aim to eliminate ecDNA, disrupt its segregation, or silence its transcription, though specificity remains a major obstacle. The central conclusion is that mutations in circular DNA are not rare biological exceptions—they are rendered invisible by a genomic paradigm that equates the genome with the chromosomes. Making these mutations visible requires nothing less than a conceptual redefinition of the genome itself.

Keywords: Extrachromosomal Circular DNA, ecDNA, eccDNA, Somatic Mutations, Genomic Instability, Cancer Evolution, Therapeutic Resistance.

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Introduction: Why Circular DNA is the Blind Spot of Genomics

The central dogma of genomics has historically been linear. From the double helix of chromosomal DNA to the readouts of next-generation sequencers, the field has been built upon a fundamental assumption: genetic information is organized, transmitted, and interpreted in a linear fashion. However, this chromosome-centric view has inadvertently rendered a significant, and highly dynamic, component of the genome invisible. Circular DNA—encompassing extrachromosomal circular DNA (eccDNA), small polydispersed circular DNA (spcDNA), and the massive oncogenic ecDNA—represents a vast, autonomous genetic universe operating in parallel to the canonical nuclear genome (Møller et al., 2018).

Unlike their linear chromosomal counterparts, these circular elements exist as mobile, non-inherited genetic entities that are present in virtually all eukaryotic cells, from healthy somatic tissue to aggressive malignancies. They are characterized by a distinct set of biophysical properties that fundamentally alter their evolutionary trajectory within a single cell cycle. Specifically, circular DNA lacks telomeres, rendering it immune to the end-replication problem but susceptible to uncontrolled rolling-circle replication. It frequently exhibits a mosaic structure, shuffling exons and regulatory elements from disparate chromosomal loci into novel combinatorial units (Paulsen et al., 2019). Most critically for the study of mutagenesis, these circles replicate with low fidelity and are subject to a distinct—and currently poorly defined—DNA damage repair (DDR) landscape.

This review posits that the failure to integrate circular DNA into standard genomic analysis constitutes a critical blind spot in contemporary genomics. The mutation rate and mutational spectrum of circular DNA are systematically underestimated by whole-genome sequencing (WGS) and whole-exome sequencing (WES), primarily because these methodologies rely on alignment to a linear reference genome. Consequently, we are currently mapping the evolution of cancer and aging while ignoring the most volatile, mutation-prone compartment of the cellular genome.

The Mutational Burden of the Invisible

The accumulation of mutations in circular DNA is not a neutral stochastic event; it is a primary driver of pathogenesis. In oncology, the focal amplification of oncogenes via ecDNA is now recognized as a ubiquitous mechanism in aggressive tumors, driving copy number heterogeneity and accelerating resistance to targeted therapies (Kim et al., 2020). However, the danger of ecDNA extends beyond copy number. Because ecDNA replicates extrachromosomally and segregates randomly during mitosis, it creates a platform for rapid Darwinian evolution. Mutations that confer a fitness advantage under therapeutic pressure can emerge and dominate a tumor population within a single cell cycle.

Furthermore, the structural configuration of circular DNA predisposes it to specific classes of mutations that are rare in linear DNA. The torsional strain associated with circularization and transcription can induce non-B DNA structures, increasing susceptibility to single-strand breaks and template-switching events during replication. This leads to the formation of complex rearrangements and "mutational mosaicism"—a phenomenon where hundreds of distinct sequence variants of a single oncogene co-exist within the same tumor (deCarvalho et al., 2018).

Beyond oncology, the accumulation of eccDNA has been implicated in cellular aging and genomic instability. The "DNA circle catastrophe" model suggests that as cells age, the extrusion and accumulation of mutagenized circular DNA from the chromosomal genome destabilizes the remaining linear DNA by titrating essential replication and repair factors (Møller et al., 2016).

The Analytical Failure of Linear Genomics

Despite the established biological relevance of these elements, they remain a blind spot due to a fundamental methodological mismatch. Standard bioinformatic pipelines for WGS and WES are optimized for detecting single nucleotide variants (SNVs) and indels against a diploid, linear template. These algorithms rely on the assumption of synteny—that the order of reads along the chromosome should match the order of the reference genome.

Circular DNA violates this assumption in three critical ways:

1. **Breakpoints**: The junction points where linear DNA was ligated into a circle produce chimeric reads that map to two distant chromosomal loci. These are routinely discarded as technical artifacts or PCR chimeras.

2. **Copy Number**: ecDNA exists in hundreds of copies per cell, leading to massive localized coverage that confounds somatic copy number alteration (SCNA) callers.

3. **Non-Reference Sequence**: eccDNA frequently contains scrambled exons and intergenic regions in non-canonical orders, which are difficult to assemble without dedicated de novo assembly strategies.

Thus, the current standard of care in genomic analysis operates under what we term the Linear Reference Bias: the systematic exclusion of sequences that do not conform to the linear architecture of the reference genome. This bias creates a dangerous vacuum of knowledge regarding the true mutational load of the cell.

Aims of this Review

This article aims to establish circular DNA as a distinct compartment of the mutational landscape. We will review the mechanisms driving the elevated mutation rate in circular versus linear DNA, quantify the "hidden" mutational burden missed by conventional pipelines, and explore the therapeutic implications of targeting mutated ecDNA. By shifting the paradigm from a static, linear genome to a dynamic, circularized one, we can begin to understand why tumors evolve so rapidly and how we might stop them.

Types of Mutations in Circular DNA

The mutational landscape of circular DNA is not a mere subset of the mutations observed in the linear nuclear genome. Rather, the biophysical configuration of covalently closed circular molecules, combined with their unique replication and segregation mechanics, gives rise to a distinct

mutational spectrum. This spectrum is characterized by an enrichment of complex structural rearrangements, a high frequency of short-range template switches, and the emergence of chimeric sequences that defy conventional karyotyping. Understanding this spectrum is essential for accurately interpreting sequencing data derived from cancerous and aging tissues (Verhaak et al., 2019).

Single Nucleotide Variants (SNVs)

While single nucleotide variants are a universal feature of genomic DNA, their etiology in circular DNA appears to be skewed toward specific sequence contexts. The replication of eccDNA and ecDNA frequently proceeds via rolling-circle mechanisms or error-prone break-induced replication (BIR). These pathways lack the processivity and fidelity of chromosomal S-phase replication. Consequently, SNVs in circular DNA are often concentrated at sites of microhomology flanking double-strand breaks.

Specifically, the repair of broken replication forks on circular templates frequently utilizes microhomology-mediated end joining (MMEJ). This error-prone repair pathway leaves distinct mutational scars, enriching for C>A and T>C transitions adjacent to the annealed microhomology tracts (Carvalho & Lupski, 2016). Furthermore, because ecDNA replicates in a "random copy number" model rather than a "once-per-cycle" model, stochastic errors during amplification become rapidly fixed in the population. A single replication error during the exponential amplification of an ecDNA molecule can generate hundreds of copies of the mutant allele within one cell cycle, effectively accelerating the fixation of otherwise rare SNVs.

Insertions and Deletions (Indels)

Short insertions and deletions are disproportionately frequent in circular DNA relative to the rest of the genome. This is primarily attributable to replication slippage on circular templates. Unlike linear chromosomes, where a stalled fork can be rescued by an adjacent origin, circular DNA replication is often initiated from a single origin. Template switching—where the nascent strand dissociates and reanneals at a downstream homologous site—is common under replicative stress.

This mechanism predominantly generates deletions, as the polymerase skips the intervening sequence (Shoshani et al., 2021). However, in cases of serial template switching, insertions can occur if the polymerase reanneals to a different eccDNA molecule or a different region of the same circle, leading to tandem duplications of short sequence blocks. The frequency of ±1 to ±50 bp indels in eccDNA isolated from aged fibroblasts has been shown to be significantly higher than in matched chromosomal DNA, suggesting that eccDNA acts as a reservoir of accumulated replication errors (Møller et al., 2018).

Structural Rearrangements

The most distinctive feature of mutated circular DNA is the prevalence of complex structural variants (SVs). Linear chromosomes suffer deletions, inversions, and translocations; circular DNA suffers rearrangements that alter its functional architecture without necessarily altering its size.

Duplications and Triplications. Due to the propensity for rolling-circle replication, eccDNA frequently evolves via intra-molecular recombination. This can result in the formation of "head-to-tail" concatemers, effectively generating triplications of oncogenic segments within a single circle. Such tandem duplications are frequently observed in ecDNA harboring EGFR and MYC amplifications (Purshouse et al., 2022).

Inversions and Fragment Shuffling. Unlike linear DNA, where an inversion requires two distant breakpoints on the same chromosome, circular DNA can invert segments via a simple intra-molecular switch in polymerase direction. This leads to the formation of palindromic structures and non-coding RNAs with novel regulatory potential.

Focal Deletions. The excision event that creates eccDNA from linear chromosomes is itself a mutagenic act. The excision of a circular molecule leaves a scar at the donor locus—a chromosomal deletion. However, once circularized, the molecule can undergo secondary deletions, removing tumor suppressor elements that were incidentally co-extruded with the oncogene (Koche et al., 2020).

Chimeric Sequences and Chromoplexy

Perhaps the most striking violation of the linear reference model is the generation of chimeric sequences within circular DNA. Single eccDNA molecules frequently contain fragments originating from two, three, or even four distinct chromosomes. This phenomenon, termed "circular DNA shuffling," occurs when multiple double-strand breaks occur across the genome, and the fragments are stitched together via non-homologous end joining (NHEJ) prior to circularization.

These events represent a form of localized chromoplexy—a "chain reaction" of rearrangements—condensed into a single extrachromosomal particle (Wu et al., 2019). The resulting chimeric circle fuses exons from unrelated genes, creating open reading frames for neo-antigens or aberrant oncogenic fusion proteins. Critically, because the constituent fragments are derived from diverse loci, these chimeric sequences are invisible to standard structural variant callers, which rely on paired-end reads mapping within a predictable distance on the same chromosome.

Epimutations: The Chromatinization Effect

Historically, eccDNA and ecDNA were considered "naked" DNA. However, recent high-resolution chromatin immunoprecipitation (ChIP) sequencing has demonstrated that large ecDNA molecules (>1 Mb) are densely chromatinized and capable of retaining histone modifications (Hung et al., 2021). This chromatinization introduces a layer of epimutational potential.

When a chromosomal fragment is excised to form a circle, it does not always retain its parental epigenetic state. Analysis of histone 3 lysine 27 trimethylation (H3K27me3) and H3K36me3 marks on ecDNA reveals discordant patterns. Circles frequently exhibit:

1. **Loss of maintenance methylation**: Passive demethylation of CpG islands due to rapid replication outpacing the maintenance methyltransferase machinery.

2. **Ectopic enhancer adoption**: A regulatory element from one chromosomal locus (e.g., a strong enhancer) physically fused to an oncogene from another locus on the same circle, leading to aberrant transcriptional activation.

These epimutations are functionally analogous to genetic mutations: they result in a heritable (though non-sequence-based) change in gene expression that confers a selective advantage.

Quantifying the Mutation Rate Differential

The cumulative effect of these mechanisms is a significant disparity in mutation rate. If we define μ as the mutation rate per base pair per replication cycle, we can express the effective mutation rate for a given oncogene (μ_g) as a function of both its point mutation rate and its copy number:

$$\mu_g = \mu_{bp} \times CN_{avg} \times f_{rep}$$

Where μ_{bp} is the base pair mutation rate, CN_{avg} is the average copy number of the circular molecule per cell, and f_{rep} is the replication frequency coefficient (estimated to be 3-5 times higher for ecDNA than for chromosomal DNA). This model explains how high-copy ecDNA can accumulate oncogenic SNVs and SVs within a clinically relevant timescale, while the same gene residing on chromosome 7 remains stable (Luebeck et al., 2023).

In summary, the mutational repertoire of circular DNA is distinct: it is structurally promiscuous, epigenetically volatile, and kinetically accelerated. Recognizing this spectrum is not merely an academic exercise; it is a prerequisite for designing therapeutic strategies that can outpace the evolution of circular genomes.

Experimental Methods for Circular DNA Isolation

The study of circular DNA mutagenesis is fundamentally constrained by the limits of detection. Unlike chromosomal DNA, which can be reliably sheared, ligated, and sequenced with minimal bias, circular DNA exists as a heterogeneous population of molecules ranging from 100 bp eccDNA to megabase-scale ecDNA. These molecules share no common sequence tag, no conserved origin of replication, and no consistent copy number. Consequently, their isolation relies not on affinity capture, but on negative selection and biophysical enrichment. This section evaluates the three principal methodological pillars of circular DNA isolation—enzymatic selection, rolling circle amplification, and conformational separation—with an emphasis on how each technique introduces specific biases that shape our understanding of the mutational landscape (Møller, 2020).

Enzymatic Selection: The Exonuclease Paradox

The most widely adopted strategy for eccDNA enrichment is the enzymatic digestion of linear DNA. This approach exploits the fundamental topological difference between open and closed DNA conformations. Linear DNA molecules possess free 5' and 3' termini, making them substrates for processive exonucleases. Circular DNA, lacking these termini, remains intact.

Two enzymes dominate this protocol. Exonuclease V (RecBCD) is a ATP-dependent helicase-nuclease that degrades linear dsDNA from both ends simultaneously. Exonuclease T is a single-stranded exonuclease that removes linear ssDNA. In combination, they provide comprehensive depletion of linear nuclear and mitochondrial fragments (Shibata et al., 2012).

Advantages and Efficiency. Under optimal conditions, exonuclease treatment achieves a 10^3 to 10^4-fold enrichment of circular over linear DNA. This permits the detection of eccDNA species present at fewer than one copy per ten cells. The protocol is rapid, scalable, and compatible with both short-read and long-read sequencing platforms.

Limitations: The Retention Artifact. However, the exonuclease method suffers from two critical limitations. First, digestion is rarely complete. Fragmented linear DNA resulting from mechanical shearing or apoptotic degradation often lacks fully unprotected ends, rendering it partially resistant to exonuclease activity. These residual linear fragments are co-sequenced and erroneously bioinformatically classified as circular junction reads, inflating estimates of eccDNA abundance (Kumar et al., 2017).

Second, and more significantly for mutagenesis studies, damaged circles are preferentially lost. Nicked circles—molecules containing single-strand breaks—are linearized during denaturation or under torsional stress. Once linearized, they become substrates for exonuclease digestion. Since nicks are often precursors to mutations or markers of active repair, this results in a systematic depletion of the very molecules most relevant to understanding circular DNA mutagenesis. The final sequencing library therefore represents a survivor bias, enriched for intact, supercoiled molecules that may not reflect the true mutational burden of the cellular circular DNA pool.

Rolling Circle Amplification (RCA)

Rolling circle amplification is not strictly an isolation method, but an amplification strategy that exploits circular topology for signal enhancement. RCA utilizes the high processivity and strand displacement activity of bacteriophage phi29 DNA polymerase. Upon encountering a circular template, phi29 continues replication indefinitely, generating a long concatemer of the circle sequence. This product can be detected by hybridization, sequenced directly, or used as template for further amplification (Dean et al., 2001).

Selectivity and Sensitivity. The principal advantage of RCA is its exquisite selectivity. Under isothermal conditions, phi29 amplifies circular templates 10,000-fold more efficiently than linear templates. This permits the detection of eccDNA from as few as 100 picograms of total DNA. For small eccDNA (<5 kb), RCA coupled with restriction digestion (Cider-seq) remains the gold standard for identifying junction sequences.

Mutational Artifacts and Amplification Bias. Despite its sensitivity, RCA introduces severe artifacts into mutational analysis. phi29 polymerase lacks 3'→5' proofreading exonuclease activity. Its error rate is approximately 1 × 10^-5 to 1 × 10^-6 per base per duplication—modest for a single pass, but catastrophic for the serial passes characteristic of RCA. A single molecule undergoing 1,000-fold amplification accumulates approximately one error per kilobase. When sequencing the resulting

concatemers, it becomes impossible to distinguish authentic somatic mutations present in the original eccDNA from polymerase errors introduced during amplification (Lou et al., 2013).

Furthermore, RCA exhibits pronounced template bias. Small circles (<1 kb) amplify with greater efficiency than large circles, due to reduced secondary structure and lower torsional strain on the polymerase. This systematically skews mutational discovery toward smaller species and underrepresents large ecDNA harboring full-length oncogenes.

Density-Gradient and Electrophoretic Separation

Prior to the advent of high-throughput sequencing, circular DNA was isolated using biophysical separation. Cesium chloride-ethidium bromide (CsCl-EtBr) density gradient centrifugation separates DNA based on supercoiling density. Intercalation of EtBr unwinds the double helix; linear and nicked circular DNA bind more EtBr than supercoiled circular DNA, reducing their buoyant density. Following ultracentrifugation, supercoiled circular DNA forms a distinct band below the linear/nicked fraction (Radloff et al., 1967).

Contemporary Applications. While CsCl gradients are laborious and low-throughput, they retain a critical advantage: they are unbiased by sequence. Unlike enzymatic selection, which may fail to excise circles lacking intact junctions, and unlike RCA, which over-represents small templates, CsCI separation preserves the native distribution of circular DNA sizes and sequences. This method is uniquely suited for the isolation of large ecDNA (>100 kb) that are poorly amplified by RCA and poorly recovered following exonuclease digestion due to their fragility.

Pulsed-field gel electrophoresis (PFGE) offers an alternative separation modality. By alternating the orientation of the electric field, PFGE resolves molecules up to several megabases in length. Circular DNA migrates anomalously compared to linear size standards, permitting the excision of gel slices specifically enriched for high molecular weight circles (Johnson & Grossman, 1977).

Limitations. These methods are severely limited by input material requirements. CsCl gradients require microgram quantities of DNA, precluding their use in liquid biopsies or single-cell applications. Additionally, extensive handling during ultracentrifugation and gel extraction introduces mechanical shearing, fragmenting large ecDNA and creating artifactual breakpoints that confound subsequent mutational analysis.

Towards Integrated Enrichment Strategies

No single method comprehensively captures the full mutational spectrum of circular DNA. Enzymatic selection remains essential for high-throughput discovery but introduces survivor bias. RCA enables single-molecule sensitivity but corrupts mutational signatures. Density separation preserves native topology but requires prohibitive input material.

Emerging strategies employ sequential enrichment: CsCl gradient isolation of total circular DNA followed by gentle, non-amplifying library preparation (e.g., nanopore sequencing of native molecules). This integrated approach minimizes both the loss of damaged circles and the

introduction of polymerase errors, offering the most accurate portrait to date of the mutational landscape of circular DNA (Wang et al., 2021).

Methods for Mutation Detection in Circular DNA

The detection of mutations in circular DNA presents a unique bioinformatic and molecular challenge. Unlike chromosomal variants, which are called against a haploid or diploid reference, mutations in eccDNA and ecDNA reside on molecules that are simultaneously highly repetitive (due to rolling-circle amplification) and highly mosaic (due to inter-locus shuffling). Conventional variant callers, designed for linear alignment, systematically misclassify circular junction reads as split-read artifacts or structural false positives. This section evaluates the three principal strategies for mutation detection in circular DNA: circle-optimized short-read sequencing, long-read single-molecule sequencing, and CRISPR-directed enrichment. Each method offers a distinct trade-off between sensitivity, specificity, and structural resolution (Zhao et al., 2022).

Circle-Optimized Sequencing (Circle-seq)

Circle-seq refers to a family of protocols that couple enzymatic enrichment of circular DNA (typically exonuclease V/T digestion) with deep short-read sequencing and junction-aware bioinformatic processing. The foundational principle of Circle-seq is not merely the enrichment of circular molecules, but the deliberate targeting of junction reads—discordant read pairs where one mate maps to a distant chromosomal locus or where the read contains a chimeric alignment spanning two non-contiguous genomic coordinates (Shibata et al., 2012).

Junction-Centric Mutation Calling. The circular junction is the most information-rich feature of any eccDNA molecule. It is the scar of the excision event, the ligation site, and frequently the locus of microhomology-mediated mutagenesis. In Circle-seq, these junctions are identified by clustering split-read alignments. True circular junctions exhibit several hallmarks: they are supported by multiple independent reads, they lack telomeric repeats, and their breakpoints are enriched for microhomology (2–6 bp) indicative of MMEJ repair.

From these junctions, the mutational landscape of the internal sequence can be inferred. By comparing the assembled circular contig to the reference genome, single nucleotide variants and small indels can be called with moderate accuracy. However, this inference is indirect. Because short-read libraries fragment the circle prior to sequencing, the physical linkage between a junction and a distal SNV is lost. It becomes impossible to determine whether two mutations reside on the same circular molecule or on distinct copies (Møller et al., 2016).

The Rare Circle Problem. The dominant limitation of Circle-seq is its poor sensitivity for low-abundance species. A typical Illumina sequencing run generates 300–500 million reads. If eccDNA constitutes 0.1% of the input DNA, and if that eccDNA population is distributed across 10,000 distinct circles, the average coverage per unique junction is below one read. Rare circles—including those harboring driver mutations in treatment-naïve tumors—fall below the

limit of detection. Deep sequencing to 1 billion reads partially mitigates this, but the relationship between sequencing depth and rare circle discovery is sublinear due to PCR duplication bias.

Long-Read Single-Molecule Sequencing

The advent of long-read sequencing platforms, principally Oxford Nanopore Technologies (ONT) and Pacific Biosciences (PacBio), has transformed the detection of structural mutations in circular DNA. Unlike short-read methods, which reconstruct circles from fragmented junction-spanning pairs, long-read sequencing can traverse the entire circular molecule in a single contiguous read (Huddleston et al., 2017).

Structural Resolution. For large ecDNA (>100 kb), long-read sequencing is not merely advantageous; it is essential. Structural mutations—including tandem duplications, inversions, and the fusion of fragments from three or more donor chromosomes—produce complex linear contigs that cannot be resolved by short-read assembly. PacBio HiFi reads (15–25 kb at 99.9% accuracy) permit the phased assembly of full-length ecDNA haplotypes. Nanopore ultra-long reads (>100 kb) can capture the entire ecDNA molecule, including its telomere-like junctional heterogeneity, in a single pass.

Error Rates and Correction. However, long-read sequencing suffers from elevated raw error rates. ONT's error profile is dominated by indels in homopolymer regions; PacBio's legacy platforms exhibited stochastic substitution errors. For mutational analysis—particularly SNV calling at single-molecule resolution—these errors are prohibitive. A raw Nanopore read with 5% error cannot distinguish a true somatic mutation from a sequencing artifact.

Contemporary solutions employ a hybrid correction strategy. High-accuracy short reads are used to polish long-read assemblies, or circular consensus sequencing (CCS) is employed, wherein the same circular molecule is sequenced multiple passes on the PacBio platform. This yields a single consensus read with accuracy exceeding Q40 (99.99%) (Wenger et al., 2019). However, CCS is length-limited; very large ecDNA cannot be circularized and resequenced.

The Methylation Axis. An ancillary advantage of Nanopore sequencing is the direct detection of DNA base modifications. Because ionic current shifts are sensitive to 5-methylcytosine and 6-methyladenine, ONT permits simultaneous genotyping and epigenotyping of single ecDNA molecules. This has revealed that ecDNA molecules bearing identical sequences frequently exhibit discordant methylation patterns, representing a form of structural epimutation invisible to short-read bisulfite sequencing (Simpson et al., 2017).

Targeted CRISPR-Capture of Circular DNA

While Circle-seq offers breadth and long-read sequencing offers structural resolution, neither method efficiently recovers specific circular loci at high depth. For investigators studying a known oncogenic ecDNA (e.g., EGFR-ecDNA in glioblastoma), the vast majority of sequencing reads are wasted on irrelevant chromosomal DNA or on eccDNA species of no biological interest.

CRISPR-Capture addresses this inefficiency. The method employs catalytically dead Cas9 (dCas9) or ribonucleoprotein complexes of active Cas9 paired with a repair-blocking oligonucleotide to bind, but not cleave, sequences flanking known circular junctions. The bound Cas9 is used as an affinity handle to pull down the entire circular molecule (Sipos et al., 2019). Alternatively, active Cas9 can be programmed to cleave immediately adjacent to a junction site, linearizing the circle and creating a uniform end for adaptor ligation.

Depth and Sensitivity. CRISPR-Capture achieves enrichment factors exceeding 1,000-fold for targeted circles. This permits the detection of minor ecDNA variants present at frequencies below 0.1% within a tumor population—a resolution unattainable by shotgun Circle-seq. Furthermore, by capturing the intact molecule, CRISPR-Capture preserves haplotype information; mutations on the same ecDNA molecule are sequenced together.

The A Priori Constraint. The method's central limitation is its requirement for sequence pre-knowledge. CRISPR guide RNAs must be designed against a known junction or a unique sequence within the circle. For discovery science in ecDNA-naïve samples, CRISPR-Capture is blind. It is therefore complementary to, rather than a replacement for, unbiased Circle-seq (Zhang et al., 2021).

Integrative Frameworks

No single detection method comprehensively resolves the mutational spectrum of circular DNA. Emerging frameworks employ a tiered strategy: unbiased Circle-seq for junction discovery, long-read sequencing for structural phasing of abundant ecDNAs, and CRISPR-Capture for deep mutational profiling of clinically-relevant resistance alleles. This integration is essential to move the field from qualitative description of eccDNA presence to quantitative assessment of eccDNA evolution.

Computational Methods for Mutation Identification

The experimental enrichment of circular DNA yields a sequencing library; however, the conversion of these reads into a catalog of somatic mutations remains a formidable computational challenge. Conventional bioinformatic pipelines are architected for a linear, diploid genome. They assume that reads should map contiguously to a reference, that coverage deviations reflect copy number alterations of linear chromosomes, and that split reads indicate germline structural variants. Circular DNA violates all three assumptions. Its reads originate from covalently closed molecules, its copy number frequently exceeds 100 per cell, and its junctions join loci that are megabases apart on the linear reference. This section examines three computational strategies essential for accurate mutational profiling of circular DNA: graph-based structural assembly, junction-aware variant calling, and copy-number-normalized burden estimation (Prada-Luengo et al., 2020).

Graph-Based Reconstruction of Circular Structures

The first computational challenge is the correct assembly of circular sequences from fragmented short-read data. Linear assembly algorithms terminate contig extension upon encountering repetitive sequence or low coverage. For circular DNA, termination is an artifact; the molecule has no end. The solution lies in graph-based genome assembly, wherein reads are represented as nodes connected by overlapping k-mers (De Bruijn graphs) or by pairwise alignments (string graphs).

Circular Contig Resolution. In a De Bruijn graph, a linear chromosome produces a path with two distinct termini—nodes of indegree or outdegree zero. A circular molecule, by contrast, produces a closed cycle. The identification of circular contigs therefore reduces to a topological problem: detecting strongly connected components without free ends (Bankevich et al., 2012). Dedicated circular DNA assemblers, such as Circle_finder and CIDER-seq's native pipeline, explicitly search for cycles in the assembly graph. When a read pair bridges the putative junction, and the graph traverses from the 3' end of the contig back to its 5' start, the molecule is classified as circular.

Overcoming Repetitive Barriers. ecDNA poses an additional challenge. These large molecules frequently contain tandem amplifications of oncogenes embedded in high-copy repeat deserts. De Bruijn graphs become tangled when k-mer multiplicity exceeds heterozygous thresholds. Recent advances employ linked-read sequencing (10x Genomics) or Hi-C proximity ligation to phase ecDNA contigs. By incorporating chromatin conformation data, assemblers can distinguish true circular ecDNA sequences from collapsed paralogous repeats (Ghareghani et al., 2022).

Junction-Aware Variant Calling

Once a circular contig is assembled, the next task is the identification of somatic mutations within it. Standard variant callers (GATK, FreeBayes) realign reads to the linear reference and penalize soft-clipping at alignment termini. For circular DNA, the true biological junction is precisely at the point of soft-clipping. Naive filtering removes these events as low-quality alignments.

Junction SNV Correction. A specific artifact plagues circular DNA variant calling. When a read spans the circular junction, the base immediately adjacent to the ligation site frequently exhibits high mismatch rates. This occurs not because of true somatic mutation, but because polymerases struggle to extend through the nick or because aligners misplace the junction by 1–2 bp. Dedicated circular DNA callers, such as Circle-Map and eccDNA_VariantCaller, implement local realignment around the junction. They realign the read as a continuous circle rather than a linear fragment, suppressing false positive SNV calls at the junction coordinate (Prada-Luengo et al., 2019).

Mosaic Mutation Detection. Circular DNA exhibits extreme mosaicism. A single tumor may harbor dozens of distinct ecDNA haplotypes, each bearing unique SNVs and indels. Conventional variant callers report variant allele frequency (VAF) relative to the total reads at that locus. For ecDNA present at 50 copies per cell, a mutation present on 5 copies yields a VAF of 10%—a signal easily detectable but often misinterpreted as subclonal chromosomal mutation. Junction-aware callers renormalize VAF to the estimated ecDNA copy number, distinguishing between low-frequency mutations in high-copy circles and clonal mutations in low-copy circles (Deshpande et al., 2019).

Copy-Number–Normalized Mutation Burden

The most pervasive conceptual error in circular DNA genomics is the conflation of mutation abundance with mutation frequency. Standard tumor mutational burden (TMB) is reported as mutations per megabase of genomic DNA, implicitly assuming one template copy per haploid genome. ecDNA violates this assumption by several orders of magnitude.

The Copy Number Problem. Consider a cell containing 100 copies of an ecDNA molecule encoding EGFR. A single replication error introduces a missense mutation into one daughter molecule. After one cell cycle, the cell contains 199 wild-type copies and 1 mutant copy. After ten cycles of random segregation and replication, the mutant allele may comprise 15% of ecDNA molecules. Sequencing this population yields 15% VAF at the mutant locus. A linear genome caller interprets this as a heterozygous subclonal mutation. The ecDNA caller recognizes that 15% VAF at copy number 100 implies 15 mutant molecules and 85 wild-type molecules per cell—a significant mutational event, yet invisible to conventional burden metrics.

The Corrected Metric. We therefore propose the adoption of a copy-number-normalized mutation burden (CNMB), defined as:

$$CNMB = (\Sigma\ VAF_i \times CN_ecDNA) / L_ecDNA$$

Where VAF_i is the variant allele frequency of mutation i within the circular compartment, CN_ecDNA is the estimated absolute copy number of the ecDNA molecule (not the chromosomal locus), and L_ecDNA is the length of the circular consensus sequence in megabases. This metric reports the average number of mutations per individual circular molecule.

Implementation. CN_ecDNA estimation itself requires computational attention. Short-read coverage depth at ecDNA loci is confounded by co-amplification of homologous chromosomal sequences. Recent tools, including AmpliconArchitect and ecSeg, integrate coverage analysis with junction density to infer ecDNA-specific copy number. By restricting coverage analysis to reads that span the unique junction or that map to regions with structural variant signatures, these tools deconvolve chromosomal and extrachromosomal amplification (Luebeck et al., 2020).

Toward Unified Computational Frameworks

The field currently suffers from computational fragmentation. Circle-seq data are processed with custom scripts; ecDNA amplicon analysis employs distinct structural variant tools; long-read circular assemblies use yet another algorithm suite. There exists no unified computational framework that accepts raw sequencing reads from any platform and outputs a fully phased, copy-number-normalized, junction-aware mutation catalog for all circular species.

Emerging efforts seek to integrate these tasks. The Circular DNA Analysis Toolkit (CDAT) and CircleBase represent initial steps toward standardization. However, these tools remain research-grade and lack the validation infrastructure required for clinical deployment (Zhao et al., 2021).

The computational challenge of circular DNA mutation detection is not merely technical; it is conceptual. As long as we treat circular DNA as a variant of linear DNA rather than a distinct genetic compartment, we will continue to miscompute its mutation rate, misattribute its evolutionary trajectory, and misestimate its contribution to disease.

Validation of Mutations in Circular DNA

The computational identification of mutations in circular DNA remains, at present, a probabilistic exercise. Short-read alignments produce junction calls with associated mapping qualities; long-read assemblies generate consensus sequences with per-base confidence scores; variant callers output SNVs with phred-scaled likelihoods. However, these in silico predictions are insufficient to establish biological truth. Circular DNA molecules are prone to PCR recombination during library preparation; their junction sequences are susceptible to misalignment from highly repetitive genomic regions; and their high copy number inflates variant allele frequencies in ways that linear callers misinterpret. Experimental validation is therefore not merely confirmatory—it is epistemologically essential (Koche et al., 2020). This section examines three validation modalities: inverse PCR for junction-specific mutation confirmation, fluorescence in situ hybridization (FISH) with mutation-specific probes for spatial localization, and single-cell circular DNA sequencing for resolution of cellular heterogeneity.

Inverse PCR: Junction-Specific Mutation Confirmation

Inverse PCR (iPCR) represents the historical gold standard for circular DNA validation. Unlike conventional PCR, which amplifies a linear fragment between two outwardly facing primers, iPCR employs primers oriented in the opposite direction—extending away from each other across the circular junction. Successful amplification occurs only when the template is circularized, producing a linear amplicon spanning the ligation site (Møller et al., 2018).

Mutation Validation at Single-Base Resolution. For mutational analysis, iPCR offers two distinct advantages. First, it provides orthogonal confirmation that the putative junction is not a bioinformatic artifact. A positive iPCR product of the expected molecular weight, Sanger sequenced across the junction, definitively establishes the precise breakpoint coordinates. Second, iPCR permits validation of SNVs and indels located within 500 bp of the junction. By designing primers flanking the variant site, the mutant allele can be amplified and sequenced from bulk DNA, confirming that the mutation is present on an extrachromosomal circular template and not merely a sequencing error or a chromosomal variant erroneously assigned to ecDNA (Shibata et al., 2012).

Limitations. iPCR is low-throughput and requires prior knowledge of junction sequence. It is suitable for validating a handful of high-priority candidates but cannot scale to the thousands of circles identified in modern Circle-seq experiments. Furthermore, iPCR from bulk DNA provides population-average data; it cannot determine what fraction of cells harbor the mutated circle, nor whether multiple mutations reside on the same physical molecule or on distinct ecDNA haplotypes.

Fluorescence In Situ Hybridization with Mutation-Specific Probes

While sequencing-based methods infer the presence of circular DNA from fragmented DNA extracts, fluorescence in situ hybridization (FISH) visualizes ecDNA molecules directly within the intact nucleus. Standard DNA FISH employs probes targeting large genomic loci (e.g., EGFR, MYC), revealing ecDNA as numerous extrachromosomal foci rather than paired chromosomal signals (Turner et al., 2017).

Mutation-Specific Discrimination. The adaptation of FISH for mutation validation requires the design of locked nucleic acid (LNA) or peptide nucleic acid (PNA) probes complementary to the mutant allele. These probes exhibit differential melting temperatures sufficient to discriminate single-nucleotide mismatches under stringent hybridization conditions. When co-hybridized with a reference probe targeting a conserved region of the same ecDNA, the spatial correlation of signals confirms that the mutant allele resides on extrachromosomal particles.

Spatial Genomics of Mutation. Mutation-specific FISH provides information unattainable by sequencing: the subnuclear distribution of mutant ecDNA. Recent studies using multiplexed FISH have demonstrated that ecDNA molecules cluster in discrete nuclear foci termed "ecDNA hubs." These hubs concentrate RNA polymerase II and transcription factors, creating microenvironments of high oncogene expression. Critically, mutant ecDNA molecules exhibit preferential clustering compared to wild-type ecDNA, suggesting that activating mutations confer not only a biochemical advantage but also a spatial segregation advantage (Hung et al., 2021).

Resolution Limits. FISH cannot resolve mutations located within highly repetitive regions or distinguish closely related ecDNA haplotypes differing by multiple SNVs. Furthermore, the spatial resolution of conventional fluorescence microscopy (200–250 nm) exceeds the diameter of small eccDNA (<1 kb, approximately 10 nm). Mutation-specific FISH is therefore primarily applicable to large ecDNA molecules (>500 kb) visible as distinct nuclear foci.

Single-Cell Circular DNA Sequencing

The most significant limitation of bulk circular DNA analysis is the loss of cellular context. A mutation detected at 5% variant allele frequency in bulk sequencing may represent either a low-frequency mutation present in many cells (each harboring a minority of mutant ecDNA copies) or a clonal mutation present in 5% of cells (each harboring exclusively mutant ecDNA). These two scenarios carry profoundly different implications for tumor evolution and therapeutic resistance. Single-cell sequencing resolves this ambiguity (Andor et al., 2020).

Technical Platforms. Single-cell circular DNA sequencing faces two formidable challenges. First, eccDNA constitutes less than 0.1% of total cellular DNA; whole-genome amplification of single cells (MALBAC, DOP-PCR, or multiple displacement amplification) must preserve this rare fraction without catastrophic allelic dropout. Second, the high GC content and repetitive architecture of ecDNA are poorly tolerated by proofreading-deficient polymerases commonly used in single-cell workflows.

Recent advances employ nanowell-based encapsulation coupled with exonuclease pre-treatment to deplete linear chromosomal DNA prior to amplification. This "scCircle-seq" protocol achieves eccDNA detection in >70% of individual cancer cells, revealing that ecDNA copy number distribution follows a heavy-tailed Pareto distribution rather than a Poisson distribution. A minority of cells function as "ecDNA reservoirs," harboring hundreds of copies, while the majority contain fewer than ten copies (Chamorro González et al., 2023).

Mutation Phasing at Single-Cell Resolution. Single-cell sequencing enables haplotype-resolved mutation phasing. By sequencing individual cells at shallow coverage (0.1–0.5×), the co-occurrence of mutations on the same ecDNA molecule can be inferred through linkage disequilibrium. Two mutations present together in the same cell more frequently than expected by chance likely reside on the same circular haplotype. This approach has revealed that ecDNA in glioblastoma evolves through punctuated bursts of mutation acquisition followed by selective sweeps, rather than gradual mutation accumulation (Luebeck et al., 2023).

Emerging Single-Molecule Platforms. The ultimate validation of circular DNA mutations is direct visualization of individual, intact molecules bearing specific sequence variants. Single-molecule imaging platforms, including DNA combing and nanochannel arrays, physically linearize DNA molecules for high-resolution fluorescence mapping. When combined with CRISPR-Cas9 labeling of specific mutant loci, these platforms permit the direct counting of mutant versus wild-type ecDNA molecules in individual cells. Early applications demonstrate that mutant ecDNA molecules exhibit a replication advantage independent of their segregation advantage, accelerating their fixation in resistant populations (Wang et al., 2021).

An Integrated Validation Hierarchy

No single validation method provides complete confirmation of circular DNA mutations. We propose a hierarchical framework: iPCR for orthogonal junction confirmation of novel circles, FISH for spatial validation of ecDNA localization in abundant populations, and single-cell sequencing for resolution of clonal architecture. This tiered approach balances throughput, resolution, and biological fidelity, transforming computational predictions into experimentally constrained biological knowledge.

Biological Interpretation: The Accelerated Landscape

The preceding sections have established that circular DNA exhibits a distinct mutational spectrum, requires specialized methods for its detection and validation, and accumulates sequence alterations at rates that exceed those of the linear chromosomal genome. However, the documentation of these phenomena is not an end in itself. The central question facing the field is one of biological interpretation: What does it mean for a cell when its circular DNA mutates? The answer challenges foundational assumptions of genomic stability, heritability, and the very definition of the genome. This section synthesizes the evidence for accelerated mutational dynamics in circular DNA and proposes a revised conceptual framework in which the genome is no longer synonymous with the chromosomes (Verhaak et al., 2019).

Accelerated Mutational Dynamics: Mechanisms and Consequences

The mutation rate of circular DNA is not merely different from that of chromosomal DNA—it is fundamentally accelerated. Several convergent mechanisms drive this acceleration. First, the replication machinery acting on circular templates is intrinsically error-prone. Rolling-circle replication, the dominant mode of ecDNA amplification, lacks the licensing controls of chromosomal S-phase replication. A single ecDNA molecule can undergo multiple replication initiation events within a single cell cycle, multiplying the opportunities for polymerase errors (Carvalho & Lupski, 2016). Second, the DNA damage response on circular templates is skewed toward error-prone repair pathways. Homologous recombination is suppressed on high-copy circles due to the difficulty of sister chromatid cohesion; microhomology-mediated end joining predominates, leaving characteristic indel scars at repair sites. Third, the physical topology of circular DNA imposes torsional stress that promotes non-B DNA conformations, increasing susceptibility to single-strand breaks and deamination events (Shoshani et al., 2021).

Quantifying the Acceleration. If we define the chromosomal mutation rate μ_{chr} as mutations per base pair per division, the effective mutation rate for a given ecDNA-encoded oncogene, μ_{eff}, must incorporate both the per-replication error rate and the increased replication frequency:

$$\mu_{eff} = \mu_{bp} \times f_{rep} \times CN_{avg}$$

Where μ_{bp} is the base substitution rate per replication event, f_{rep} is the relative replication frequency of ecDNA compared to chromosomal DNA (empirically estimated at 3–5), and CN_{avg} is the average copy number of the ecDNA molecule. For a highly amplified ecDNA present at 50 copies per cell, μ_{eff} exceeds μ_{chr} by two orders of magnitude (Luebeck et al., 2023). This acceleration has profound biological consequences. A tumor cell harboring amplified ecDNA does not wait months or years to acquire resistance-conferring mutations; it generates them continuously, in every cell cycle, across hundreds of template molecules.

Non-Heritable Mutations and Functional Phenotypes

The accelerated mutation rate of circular DNA produces a vast repertoire of sequence variants. However, the relationship between mutation and phenotype in the circular compartment is complicated by a critical factor: non-heritability.

Chromosomal mutations, once fixed, are inherited by all descendant cells. They represent permanent, cumulative alterations to the germline or somatic lineage. Circular DNA mutations follow different rules. During mitosis, ecDNA molecules segregate randomly to daughter cells. A mutation present on one ecDNA copy in a mother cell may be lost entirely if that specific molecule fails to replicate or segregates to the wrong daughter. Conversely, a mutation present on a single ecDNA molecule can, through preferential amplification, come to dominate the ecDNA population within a few cell generations (deCarvalho et al., 2018).

Transient Dominance, Lasting Effect. This stochastic inheritance pattern produces a phenomenon we term transient mutational dominance. A highly advantageous mutation arising on a single

ecDNA molecule can rapidly amplify to high copy number, drive high levels of oncogene expression, and confer a profound fitness advantage—all while remaining entirely absent from the chromosomal genome. The mutation dominates the cellular phenotype without ever integrating into the stable genetic archive. If the selective pressure is removed (e.g., upon drug withdrawal), the mutant ecDNA population may contract or be lost, leaving no trace of its former dominance (Nathanson et al., 2014).

This temporal decoupling of genotype and phenotype has profound implications for precision oncology. Biopsies acquired at diagnosis may lack resistance mutations that exist only transiently during treatment. Liquid biopsies detecting circulating ecDNA may reveal mutant alleles that are absent from contemporaneous tissue biopsies. The circular genome operates on a timescale of days, not years.

Beyond the Chromosome-Centric Paradigm

The cumulative evidence regarding circular DNA mutagenesis necessitates a fundamental revision of how we conceptualize the genome. The dominant paradigm, inherited from classical genetics and molecular biology, equates the genome with the chromosomal complement. The human genome is defined as the sum of sequences on chromosomes 1–22, X, and Y. Everything else—mitochondrial DNA, plasmids, viruses, eccDNA—is categorized as "extragenomic," "extra chromosomal," or simply omitted from reference assemblies.

This chromosome-centric model is no longer tenable. It fails to account for the quantitative dominance of ecDNA in aggressive cancers, where a single ecDNA molecule may carry more transcriptional output of an oncogene than both chromosomal alleles combined. It fails to account for the evolutionary primacy of circular DNA, which generates adaptive mutations faster than the chromosomal genome can fix neutral polymorphisms. And it fails to account for the functional integration of circular and linear compartments, where chromosomal fragments are excised to form eccDNA, mutate extensively, and occasionally reintegrate into chromosomes, seeding new structural variants (Paulsen et al., 2019).

A New Definition. We propose that the genome be redefined as the complete set of heritable and non-heritable DNA molecules within a cell that contribute to its phenotypic state. This definition encompasses chromosomal DNA, mitochondrial DNA, and—critically—the dynamic population of circular DNAs that fluctuate in copy number, mutate rapidly, and modulate gene expression independently of chromosomal regulation. The genome is not a static sequence printed in the reference assembly; it is an ecosystem of interacting linear and circular replicons.

Implications for Evolution and Medicine

Adopting this expanded genomic framework transforms our understanding of somatic evolution. Chromosomal mutations provide the long-term archive of evolutionary history, accumulating steadily over decades. Circular DNA mutations provide the short-term adaptive toolkit, enabling rapid responses to acute stressors. The two systems operate on different timescales, under different error rates, and with different inheritance rules. Together, they constitute a dual-speed genome: a

slow, stable chromosomal component ensuring intergenerational continuity, and a fast, volatile circular component enabling intra-generational adaptation (Wu et al., 2019).

Therapeutic Implications. This dual-speed model carries direct therapeutic implications. Targeting the replication machinery of ecDNA—for example, through inhibition of rolling-circle initiation or disruption of ecDNA segregation—may be therapeutically advantageous even if it does not eliminate every ecDNA molecule. Slowing the mutation rate of the circular compartment could extend the duration of response to targeted therapies by delaying the emergence of resistance mutations. Conversely, therapies that inadvertently accelerate ecDNA mutagenesis (e.g., certain DNA-damaging chemotherapies) may hasten the evolution of resistant clones (Kim et al., 2020).

Conclusion of Synthesis

The circular DNA compartment is not a passive reservoir of genomic waste. It is an active, accelerated evolutionary laboratory operating in parallel to the stable chromosomal archive. Its mutations are more frequent, its selection is more rapid, and its phenotypic effects are dissociable from stable inheritance. To understand cancer evolution, aging, and acquired drug resistance, we must abandon the assumption that the genome is synonymous with the chromosomes. The genome circles, and in circling, it mutates.

Limitations and Unresolved Questions

The preceding sections have presented a coherent narrative: circular DNA is a distinct genomic compartment with an elevated mutation rate, a unique mutational spectrum, and profound biological consequences for cancer evolution and cellular aging. However, this narrative rests on methodological and conceptual foundations that remain incompletely validated. The field of circular DNA genomics is approximately fifteen years old—young by the standards of human genetics. Its methods are heterogeneous, its reference frameworks are absent, and its clinical translation is hindered by a lack of standardization. This section critically examines the major limitations and unresolved questions that constrain current understanding and proposes priority areas for method development (Haber & Schimke, 2021).

The Absence of a Universal Circular Reference Genome

The most fundamental obstacle to accurate mutation detection in circular DNA is the absence of a canonical reference. Chromosomal variant calling relies on the existence of a high-quality, population-averaged reference genome (GRCh38/hg38). Every read is aligned to this fixed coordinate system; deviations from the reference are reported as variants. Circular DNA has no such reference. Each eccDNA molecule is a unique, somatically generated fusion of chromosomal fragments arranged in an order that does not exist in the linear reference. There is no "wild-type" eccDNA sequence against which to call mutations (Dutta & Sengupta, 2022).

Consequences for Mutation Calling. The absence of a circular reference forces investigators to adopt one of two suboptimal strategies. The first strategy aligns reads to the linear reference

genome and infers circular structure from discordant alignments. This approach is computationally efficient but systematically fails to detect mutations within the circular sequence that do not coincide with the junction. The second strategy performs de novo assembly of circular contigs from enriched sequencing data, then aligns these contigs back to the linear reference to identify rearrangements and sequence variants. This approach preserves internal sequence information but introduces assembly errors, particularly in repetitive regions and at sites of high GC content (Prada-Luengo et al., 2020).

The Path Forward. A community resource analogous to the reference genome—a database of recurrent eccDNA junctions and their associated sequence variants—is urgently needed. Such a resource would not replace sample-specific assembly but would provide a standardized coordinate system for reporting mutations. Early efforts, including CircleBase and eccDB, represent initial steps, but these databases remain incomplete and are biased toward cancer-derived samples (Zhao et al., 2021). A human circular DNA reference, stratified by tissue type and disease state, is a prerequisite for reproducible mutation calling.

Distinguishing True Mutations from Artifacts

The problem of artifact discrimination pervades every stage of circular DNA analysis. Three distinct sources of error confound mutational interpretation.

Amplification Artifacts. Rolling circle amplification (RCA) and multiple displacement amplification (MDA) are essential for detecting low-abundance eccDNA species. However, phi29 polymerase lacks proofreading activity and exhibits error rates of approximately 1×10^{-5} to 1×10^{-6} per base per duplication. A circle amplified 1,000-fold accumulates, on average, one polymerase error per kilobase. When these amplified products are sequenced, authentic somatic mutations and in vitro replication errors are indistinguishable (Lou et al., 2013). The field currently lacks an experimental or computational method to confidently discriminate between these classes of variants.

Replication Errors. Even in the absence of in vitro amplification, circular DNA isolated from cells contains a high burden of replication errors. Rolling-circle replication in vivo is intrinsically error-prone. A variant detected at 5% allele frequency in a sequencing library may represent a genuine somatic mutation present in a subset of circular molecules, or it may represent a replication error that occurred during the final cell cycle prior to DNA extraction. Distinguishing between these possibilities requires longitudinal sampling or single-molecule lineage tracing—methods not yet adapted for circular DNA (Shoshani et al., 2021).

Junction Misalignment. The junction-spanning reads that define circular DNA are also sources of false positive variant calls. When a read traverses the ligation site, the aligner must split the read into two segments mapping to distant genomic loci. The exact breakpoint coordinate is ambiguous at single-nucleotide resolution. Many variant callers report mismatches at these breakpoints as SNVs, when in fact the apparent mismatch is an alignment artifact arising from microhomology at the excision site. Dedicated circular DNA callers partially mitigate this problem, but no current algorithm achieves perfect specificity (Deshpande et al., 2019).

The Challenge of Rare Circles

The mutational landscape of circular DNA is dominated by a power-law distribution: a small number of ecDNA species exist at high copy number and dominate sequencing libraries; a vast number of eccDNA species exist at extremely low abundance and are detected only sporadically. This long tail of rare circles presents a formidable detection problem.

For a circle present at one copy per 100 cells, conventional Circle-seq at 500 million read depth yields, on average, fewer than one junction-spanning read. Increasing sequencing depth to 2 billion reads improves detection but encounters diminishing returns due to PCR duplication and the finite complexity of Illumina libraries. Rare circles—including those harboring driver mutations in early-stage tumors or pre-neoplastic lesions—remain systematically under-sampled (Møller et al., 2018).

Single-Cell Resolution. Single-cell sequencing offers a potential solution but introduces new challenges. The eccDNA content of an individual cell is approximately 0.01–0.1% of its total DNA. Whole-genome amplification of single cells exhibits pronounced allelic dropout and GC bias; rare eccDNA species are frequently lost entirely. Recent nanowell-based methods achieve eccDNA detection in 70% of cancer cells, but sensitivity for circles present at fewer than five copies per cell remains poor (Chamorro González et al., 2023).

The Absence of Clinical Standards

The translational promise of circular DNA analysis—as a biomarker, a prognostic indicator, and a therapeutic target—cannot be realized without standardization. Currently, no two research groups employ identical protocols for eccDNA isolation, library preparation, sequencing, or bioinformatic analysis. Exonuclease digestion conditions vary in enzyme concentration, incubation time, and temperature. RCA amplification times range from 2 to 16 hours. Sequencing platforms include Illumina, MGI, Nanopore, and PacBio. Bioinformatic pipelines are custom-written and rarely benchmarked against ground-truth datasets (Zhao et al., 2022).

The Reference Material Problem. Clinical genomics relies on well-characterized reference standards (e.g., NA12878, HapMap samples) to validate assay performance. No such reference exists for circular DNA. A human cell line with a stably characterized ecDNA population, at defined copy number and with known junction sequences and mutation status, is urgently needed. Such a reagent would enable cross-laboratory validation, method optimization, and regulatory approval for diagnostic applications.

Interpretation Guidelines. Even when mutations are confidently identified, their clinical interpretation remains ambiguous. A KRAS G12D mutation on chromosomal DNA is a well-characterized oncogenic driver. The same mutation on an ecDNA molecule, present at 50 copies per cell in 10% of tumor cells, has unknown significance. Does it confer resistance? Does it predict poor outcome? Should it influence therapeutic selection? The field currently lacks the longitudinal clinical datasets required to answer these questions (Verhaak et al., 2019).

Priority Areas for Method Development

Addressing these limitations requires coordinated effort across four domains. First, the development of amplification-free circular DNA sequencing methods to eliminate polymerase-induced artifacts. Second, the creation of public, curated reference databases of recurrent circular DNA junctions and their associated variants. Third, the generation and distribution of certified reference materials for assay validation. Fourth, prospective clinical trials correlating circular DNA mutation status with patient outcomes.

The study of circular DNA mutagenesis has advanced remarkably from its origins in viral replication and plasmid biology. However, the transition from a descriptive science to a predictive, clinically actionable discipline requires acknowledging the limitations of current methods and investing rigorously in their resolution.

Future Perspectives

The preceding sections have established that circular DNA is not a biological curiosity but a central component of the somatic genetic landscape. Its mutation rate is accelerated, its structural plasticity is profound, and its phenotypic consequences—from oncogene amplification to drug resistance—are clinically measurable. Yet the field remains in its infancy. The methods described in this review are primarily research tools, confined to specialized laboratories and absent from routine clinical genomics. The unresolved questions outlined in Section 8 are not merely technical nuisances; they are barriers to translation. This final section projects forward, examining how the analysis of circular DNA mutations might transition from the benchtop to the bedside. Three horizons are considered: the integration of circDNA analysis into oncogenomic panels, the use of circDNA mutations as biomarkers of aging and cellular stress, and the therapeutic targeting of ecDNA as a distinct axis of cancer treatment (Haber & Schimke, 2021).

Integration into Oncogenomic Panels

Clinical cancer genomics has matured rapidly over the past decade. Targeted sequencing panels (e.g., MSK-IMPACT, FoundationOne) now routinely profile hundreds of cancer-associated genes from formalin-fixed, paraffin-embedded tissues. Circulating tumor DNA (ctDNA) assays detect actionable mutations from plasma with high sensitivity. Conspicuously absent from these panels is any assessment of circular DNA. ecDNA amplifications—among the most potent drivers of oncogene expression and therapeutic resistance—are not reported by standard copy number algorithms. Mutations residing exclusively on ecDNA are invisible to exon-only panels that align reads to the linear reference genome (Cheng et al., 2021).

The Case for Inclusion. The inclusion of ecDNA analysis in clinical panels is justified by three lines of evidence. First, ecDNA is not rare. Reanalysis of The Cancer Genome Atlas (TCGA) using ecDNA-aware algorithms has demonstrated that extrachromosomal amplification of EGFR, MYC, CDK4, and other oncogenes is present in approximately 14% of all tumors and exceeds 40% in glioblastoma and liposarcoma (Kim et al., 2020). Second, ecDNA status is prognostic. Patients with

ecDNA-positive tumors exhibit significantly shorter progression-free and overall survival than patients with chromosomal amplification of the same oncogenes. Third, ecDNA status predicts therapeutic resistance. Longitudinal sampling of patients receiving EGFR inhibitors reveals that resistance-conferring mutations (e.g., EGFR T790M, EGFR C797S) frequently arise on ecDNA templates before they are detectable in chromosomal DNA (Nathanson et al., 2014).

Implementation Pathways. Integrating ecDNA analysis into existing clinical workflows does not require novel sequencing. The primary data—short-read Illumina sequences from tumor-normal paired samples—are already generated. What is required is the deployment of ecDNA-aware bioinformatic pipelines that detect focal amplifications with junction-spanning reads and discriminate extrachromosomal from chromosomal copy number gains. Several such pipelines (AmpliconArchitect, ecSeg, Circle-Map) are validated for research use; their adaptation to Clinical Laboratory Improvement Amendments (CLIA) standards is a priority (Luebeck et al., 2020).

Circular DNA Mutations as Biomarkers

Beyond oncology, the mutational load of circular DNA may serve as a quantitative biomarker of cellular health. Unlike chromosomal mutations, which accumulate slowly over decades and require cell division for fixation, eccDNA mutations arise rapidly in response to acute stressors and can be detected from minute quantities of input DNA.

Markers of Aging. The "DNA circle catastrophe" hypothesis posits that aging cells accumulate eccDNA species extruded from unstable genomic regions. Cross-sectional studies of human fibroblasts have demonstrated a positive correlation between donor age and eccDNA abundance, as well as an increased burden of indels within eccDNA sequences (Møller et al., 2018). Longitudinal studies in model organisms confirm that eccDNA mutation burden increases monotonically with replicative age. Whether eccDNA mutation load predicts chronological age more accurately than epigenetic clocks, and whether it correlates with functional decline, remains to be determined.

Indicators of Cellular Stress. Genotoxic stress—from radiation, chemotherapy, or reactive oxygen species—induces double-strand breaks. These breaks are substrates for eccDNA formation. Preclinical models demonstrate that ionizing radiation increases eccDNA abundance by an order of magnitude within 24 hours, with the resulting circles bearing distinct mutational signatures of NHEJ repair. Peripheral blood eccDNA quantification may therefore serve as a pharmacodynamic biomarker of DNA damage, enabling real-time monitoring of chemotherapy-induced genotoxicity (Kumar et al., 2017).

Early Signals of Oncogenic Transformation. The transition from normal tissue to carcinoma in situ is accompanied by genomic instability. eccDNA excision is an early manifestation of this instability. In Barrett's esophagus, eccDNA species harboring MYC and EGFR sequences are detectable years before histopathological evidence of high-grade dysplasia (Luebeck et al., 2023). Similarly, in pre-neoplastic liver tissue from hepatitis C patients, eccDNA junctional complexity correlates with cirrhosis severity and predicts hepatocellular carcinoma risk. The development of liquid biopsy assays targeting cancer-associated eccDNA junctions may enable earlier detection of

transformation than current mutation-based ctDNA assays, which require pre-existing driver mutations.

Therapeutic Targeting of ecDNA

The most ambitious perspective—and the one with the greatest potential clinical impact—is the deliberate therapeutic targeting of ecDNA. If ecDNA is the engine of oncogene amplification and accelerated mutation acquisition, disabling that engine should attenuate tumor fitness and delay resistance.

Conceptual Strategies. Four distinct therapeutic strategies are under investigation. First, ecDNA elimination: small molecules or biologics that induce selective degradation of extrachromosomal circles while sparing chromosomal DNA. Preclinical studies demonstrate that bromodomain inhibitors (JQ1) and certain DNA intercalators reduce ecDNA copy number in glioblastoma models, although the mechanism remains incompletely understood (Wu et al., 2019). Second, segregation disruption: agents that interfere with the random segregation of ecDNA during mitosis, forcing ecDNA molecules into a single daughter cell and reducing population-level copy number. Third, replication inhibition: selective targeting of the rolling-circle replication machinery that maintains ecDNA copy number. The identity of the polymerase(s) responsible for ecDNA replication in vivo remains unknown, representing a critical gap in target identification. Fourth, transcriptional interference: disruption of the ecDNA hubs that concentrate RNA polymerase II and drive high oncogene expression (Hung et al., 2021).

The Challenge of Specificity. The central challenge for ecDNA-directed therapy is specificity. ecDNA is DNA; agents that damage or eliminate ecDNA will also damage chromosomal DNA. The therapeutic window depends on differential dependence: cancer cells may be addicted to ecDNA-encoded oncogenes, whereas normal cells lack ecDNA entirely. Identifying chemical matter with selective activity against extrachromosomal circles—or against the ecDNA-encoded transcriptome—is a priority for chemical biology.

A Convergent Trajectory

The three perspectives outlined above are not independent. The integration of ecDNA analysis into diagnostic panels will identify patient populations most likely to benefit from ecDNA-directed therapies. The validation of eccDNA mutation burden as a biomarker of aging and stress will enable preventive monitoring of at-risk populations. The development of ecDNA-targeted therapeutics will provide interventions for patients identified through these screening programs.

The study of circular DNA mutagenesis has progressed from descriptive observation to mechanistic understanding to translational application. The next decade will determine whether this trajectory continues to the ultimate destination: improved outcomes for patients with ecDNA-driven cancers and, perhaps, interventions that slow the mutational clock of aging itself.

Key Conclusion: Mutations in Circular DNA Are Not Rare — They Are Simply Invisible to Linear Genomics

The central argument of this review is not that circular DNA is a newly discovered genetic entity. Extrachromosomal circular DNA has been observed in eukaryotic cells since the 1960s, first in electron micrographs of rapidly dividing plant cells and later in mammalian tissues (Radloff et al., 1967; Cox & Lerman, 1969). Nor is the argument that circular DNA is quantitatively insignificant. In some aggressive cancers, ecDNA constitutes the majority of transcriptionally active oncogene templates, and its copy number exceeds that of its chromosomal locus by orders of magnitude (Turner et al., 2017). Rather, the central thesis is that circular DNA mutations are systematically and structurally excluded from contemporary genomic analysis. They are not rare; they are rendered invisible by a methodological paradigm that privileges linearity, diploidy, and chromosomal inheritance.

The Architecture of Invisibility

The invisibility of circular DNA mutations is not a failure of technology but a consequence of a priori assumptions embedded in the design of genomic workflows. These assumptions operate at every level of analysis.

At the molecular level, DNA extraction protocols are optimized for high molecular weight linear DNA. Circular DNA, particularly small eccDNA molecules (<1 kb), is inefficiently precipitated and frequently lost during spin column purification. When retained, it co-migrates with fragmented linear DNA and is often excluded from size selection steps that favor 300–500 bp fragments for Illumina library preparation (Møller et al., 2018).

At the enrichment level, exome capture probes are designed against reference chromosomal sequences. ecDNA molecules containing rearranged intron–exon boundaries or chimeric fusions from multiple chromosomal loci hybridize inefficiently or not at all. A mutation residing on an ecDNA molecule that fuses the EGFR kinase domain to a cryptic enhancer from chromosome 7 intergenic region will not be captured by standard exome baits, regardless of its clinical significance (Koche et al., 2020).

At the sequencing level, short-read platforms generate fragments that are, by design, shorter than the circular molecules they seek to characterize. The physical linkage between a junction and a distal SNV is destroyed during fragmentation. Two mutations present on the same ecDNA molecule appear as independent events, their cooperative selective advantage obscured (Deshpande et al., 2019).

At the bioinformatic level, alignment to the linear reference genome is the foundational operation of modern genomics. Reads that map discontinuously—split between two distant chromosomal loci—are flagged as anomalies. Chimeric alignments are filtered, soft-clipped bases are ignored, and discordant read pairs are set aside for structural variant analysis, where they are often discarded as low-confidence calls. The very signal that defines circular DNA is systematically subtracted as noise (Prada-Luengo et al., 2019).

At the interpretive level, clinical genomics reports variant allele frequencies normalized to diploid copy number. An ecDNA-encoded mutation present at 20% VAF on a molecule with 50 copies per cell is annotated as a subclonal event of uncertain significance. The interpreter never learns that this mutation resides on 10 copies of ecDNA per cell—a massive expansion of a mutant allele—because the copy number of the circular template is not measured and cannot be inferred from linear alignment (Luebeck et al., 2023).

The Consequence of Invisibility

What is lost when circular DNA mutations are rendered invisible? The empirical record of the past decade provides an answer: we lose the ability to understand the fastest, most adaptive, and most therapeutically relevant compartment of the cancer genome.

We lose prognostic discrimination. Patients with ecDNA-amplified EGFR survive a median of 12 months; patients with chromosomal EGFR amplification survive 24 months. This distinction—more powerful than many FDA-approved biomarkers—is not reported by standard genomic assays (Kim et al., 2020).

We lose mechanistic understanding. The acquisition of resistance to osimertinib in EGFR-mutant lung cancer occurs through the emergence of EGFR C797S. Serial sampling of patients reveals that this mutation frequently arises on ecDNA templates, amplifies rapidly under drug pressure, and recedes upon drug withdrawal. This dynamic, reversible resistance mechanism is invisible to single-timepoint chromosomal sequencing (Nathanson et al., 2014).

We lose therapeutic opportunity. Preclinical models demonstrate that ecDNA can be selectively eliminated from tumor cells using bromodomain inhibitors and certain DNA intercalators. These agents reduce ecDNA copy number, suppress oncogene expression, and restore drug sensitivity. Yet clinical trials of these agents do not stratify patients by ecDNA status, and correlative biomarkers of ecDNA response are not collected. A therapeutic axis remains unexplored because the patients most likely to benefit cannot be identified (Wu et al., 2019).

A New Visibility

Making circular DNA mutations visible requires more than the addition of a new bioinformatic module to existing pipelines. It requires a conceptual reorientation: the recognition that the genome is not coextensive with the chromosomes. The genome is an ecosystem of interacting replicons—linear and circular, chromosomal and extrachromosomal, stable and mobile—that together determine the phenotypic state of the cell.

This reorientation has practical implications. First, reference genomes must accommodate circularity. This does not mean abandoning GRCh38, but supplementing it with databases of recurrent eccDNA junctions and ecDNA structural haplotypes. A mutation should be reported not only by its chromosomal coordinate but also by its circular context. Second, variant callers must become topology-aware. The assumption that reads should align contiguously to a single linear locus must be replaced by probabilistic models that accommodate circular alignment, chimeric

fusion, and extrachromosomal copy number. Third, clinical reporting must adopt copy-number-normalized metrics. Reporting the mutation burden per circular molecule, rather than per diploid genome, would transform our understanding of ecDNA-driven resistance.

The Mutation That Was Always There

Consider a final thought experiment. A tumor biopsy is sequenced. The bioinformatic pipeline reports no actionable mutations. The patient receives standard chemotherapy and progresses rapidly. Autopsy reveals widespread metastatic disease. Retrospective analysis using ecDNA-aware methods identifies an EGFR C797S mutation on a highly amplified ecDNA molecule—present at the time of initial biopsy, invisible to the original analysis.

This scenario is not hypothetical. It has occurred, and continues to occur, in research cohorts reanalyzed with circular DNA-sensitive methods (deCarvalho et al., 2018). The mutation was not absent; it was not rare; it was not below the limit of detection. It was simply incompatible with the linear reference model.

The central claim of this review is that circular DNA mutations are not exotic variants confined to a small subset of tumors. They are a fundamental feature of the somatic genetic landscape, present across cancer types, aging tissues, and stressed cells. Their apparent rarity is an artifact of the methods used to observe them.

Circles that mutate are not exceptions to the rule of genomic stability. They are the rule itself—the manifestation of a dynamic, responsive, and evolutionarily adaptable genome. For decades, the field of genomics has been mapping the stars while ignoring the planets. It is time to turn the telescope toward the circular darkness and recognize that what we have dismissed as empty space is, in fact, crowded with circles, teeming with mutations, and urgent with clinical consequence.

The mutations are not rare. They are waiting to be seen.

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